

Gen. Functional for Connected Green's f^n and the Effective Action:

Free energy: $W[\mathcal{J}]$

W : generating f^n of Connected Green f^n s

$\Gamma[\phi]$: Effective Action

Γ : Gen. f^n of the 1pt Irreducible Green f^n

Claim: $Z[\mathcal{J}] = e^{-W[\mathcal{J}]}$

$$\delta^n Z[\mathcal{J}]$$

$$[\mathcal{J}] \equiv Z[\mathcal{J}]$$

$$G_n(x_1, \dots, x_n)_\mathcal{J} \equiv \delta \mathcal{J}_1 \delta \mathcal{J}_2 \dots \delta \mathcal{J}_n$$

$$\hookrightarrow \equiv \boxed{\mathcal{J}} \begin{matrix} \bullet x_1 \\ \vdots \\ \bullet x_n \end{matrix} \equiv G(x_1, \dots, x_n)_\mathcal{J}$$

$[\mathcal{J}]$: zero pt f^n $Z[\mathcal{J}] \xrightarrow{\bullet} x = \Delta_{ij} \mathcal{J}_j$

$\boxed{\mathcal{J}} \bullet : G(x)_\mathcal{J}$

ex: 1pt f^n $G(x)_\mathcal{J}$ in ϕ^4 :

$$G(x)_\mathcal{J} = \int d^d y \Delta(x \rightarrow y) \mathcal{J}(y) - \mathcal{J} \int \int \int \int \Delta \dots$$

$$\bullet \boxed{\mathcal{J}} = \begin{matrix} \bullet x \\ \bullet x \end{matrix} \xrightarrow{x} + \begin{matrix} \bullet x \\ \bullet x \end{matrix} \xrightarrow{x} \begin{matrix} \bullet x \\ \bullet x \end{matrix} + \begin{matrix} \bullet x \\ \bullet x \end{matrix} \xrightarrow{x} \begin{matrix} \bullet x \\ \bullet x \end{matrix} + \dots$$

Proof.

$$Z[\mathcal{J}] = e^{-\int d^d x \frac{\delta \mathcal{L}}{\delta \mathcal{J}} \mathcal{J}} e^{\mathcal{J} \Delta \mathcal{J}}$$

$$= (1 - \lambda \frac{\delta^4}{\delta \mathcal{J}^4}) (e^{\mathcal{J} \Delta \mathcal{J}}) |_{\mathcal{J}=0}$$

$$= \frac{\delta^4}{\delta \mathcal{J}^4} e^{\mathcal{J} \Delta \mathcal{J}}$$

$$= \delta \mathcal{J}_x^3 (\Delta_{xw} \mathcal{J}_w) e^{\mathcal{J} \Delta \mathcal{J}}$$

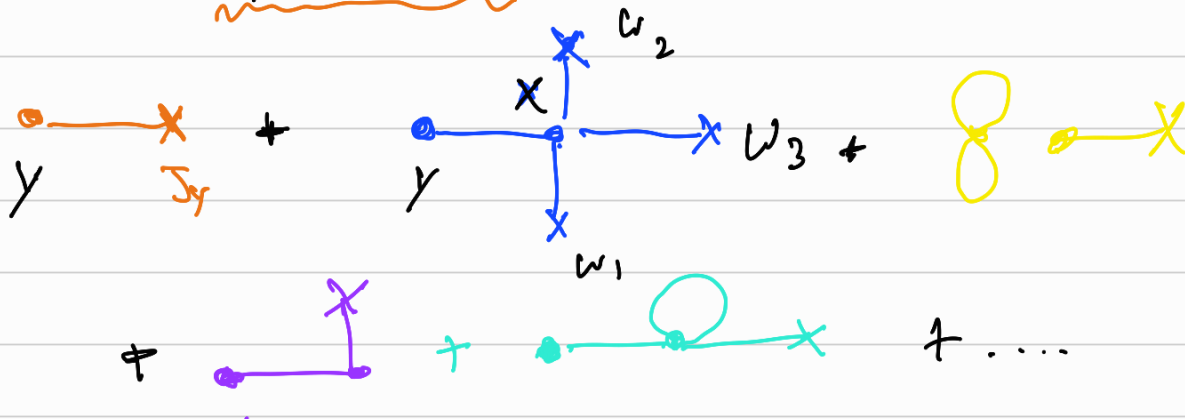
$$= \delta \mathcal{J}_x^2 (\Delta_{xx} e^{\mathcal{J} \Delta \mathcal{J}} + (\Delta_{xw} \mathcal{J}_w)^2 e^{\mathcal{J} \Delta \mathcal{J}})$$

$$= \delta \mathcal{J}_x (\Delta_{xx} \Delta_{xw} \mathcal{J}_w e^{\mathcal{J} \Delta \mathcal{J}} + 2 \Delta_{xw} \mathcal{J}_w \Delta_{xx} e^{\mathcal{J} \Delta \mathcal{J}} + (\Delta_{xw} \mathcal{J}_w)^3 e^{\mathcal{J} \Delta \mathcal{J}})$$

$$= 3 \Delta_{xx}^2 e^{\mathcal{J} \Delta \mathcal{J}} + 3 \Delta_{xx} (\Delta_{xw} \mathcal{J}_w)^2 e^{\mathcal{J} \Delta \mathcal{J}}$$

$$+ 3 (\Delta_{xw} \mathcal{J}_w)^2 e^{\mathcal{J} \Delta \mathcal{J}} + (\Delta_{xw} \mathcal{J}_w)^4 e^{\mathcal{J} \Delta \mathcal{J}}$$

$$G_{yy} = -i \left(3 \Delta_{xx}^2 (\Delta_{yw} \mathcal{T}^w) + 12 (\Delta_{xw} \mathcal{T}^w) \Delta_{xy} e^{\mathcal{T}\Delta\mathcal{J}} \right. \\
+ 6 \Delta_{xx} \Delta_{xy} \Delta_{xw} \mathcal{T}^w e^{\mathcal{T}\Delta\mathcal{J}} \\
+ 4 \Delta_{xy} (\Delta_{xw} \mathcal{T}^w)^3 e^{\mathcal{T}\Delta\mathcal{J}} + (\Delta_{xw} \mathcal{T}^w)^4 \Delta_{xy} e^{\mathcal{T}\Delta\mathcal{J}} \left. + \Delta_{yw} \mathcal{T}^w e^{\mathcal{T}\Delta\mathcal{J}} \right)$$



We should also have

$$\text{diagram} = \alpha \Delta_{ya} \Delta_{ab}^3 \Delta_{bw} \mathcal{T}^w$$

connected

$$\text{diagram} = \text{connected} \times \text{diagram}$$

$$\delta\mathcal{J}/\delta\mathcal{J} = -\delta\mathcal{W}/\delta\mathcal{J} Z[\mathcal{J}]$$

$$\text{Recall } Z[\mathcal{J}] = e^{\text{connected}} e^{\text{bubbles}}$$

solving for Z in terms of $\mathcal{W}[\mathcal{J}]$

$$Z[\mathcal{J}] = N e^{-\mathcal{W}[\mathcal{J}]}$$

Effective Action & 1PI Green's functions:

ex. of 1PI diagram:

1P Reducible:

In thermo we find that

$$G = F - Q\Phi$$

* Need to find the object conjugate to J

Ex: the electrical potential Φ is conjugate to Q in G .

* The Classical Field $\phi_c[J]$ is conjugate to J

$$\begin{aligned}\phi_c[J] &\equiv \frac{\langle 0|\hat{\phi}(x)|0\rangle_J}{\langle 0|0\rangle_J} = \frac{1}{Z[J]} \int D\phi e^{-S[\phi] + J \cdot \phi} \\ \phi(x) &= \frac{1}{Z[J]} \frac{\delta Z}{\delta J_x} \\ &= \frac{\delta}{\delta J_x} (-W[J]) = G_1(x; J)\end{aligned}$$

example: (Free scalar field in discretized theory)

$$\begin{aligned}S_D - J \cdot \phi &= \frac{1}{2} \int d^d x [(D_\mu \phi)^2 + m^2 \phi^2] - \int d^d x J(x) \phi(x) \\ &\equiv \frac{1}{2} \phi_i \Delta_{ij}^{-1} \phi_j - J_n \phi^n\end{aligned}$$

* Here we have the classical E.O.M:

$$\frac{\delta S_0}{\delta \phi^i} = \Delta_{ij}^{-1} \phi_j - J_i = 0$$

$\therefore$

$$\boxed{\phi_i^{cl} = \Delta_{ij} J_j} \quad (\text{Coulombs law})$$

$$\boxed{\phi(x) = \int d^d y \Delta(x, y) J(y)}$$

* Recall the free Partition f^N is given by

$$Z_0[J] = \exp\left[\frac{1}{2} J \Delta J\right]$$

$$\text{and since } Z_0[J] = e^{-W_0[J]}$$

this implies

$$W_0[J] = -\frac{1}{2} J \Delta J$$

and so

$$\phi_i^{cl,0} = \frac{\delta}{\delta J_i} (-W_0[J]) = \Delta_{ij} J_j$$

* The classical field



Thus, a classical field is

$$\phi^{cl} = \frac{\langle 0 | \hat{\phi} | 0 \rangle_S}{\langle 0 | 0 \rangle_S}$$

(J): Connected
Prices

2PI Green's f^{uL}:

$$i \text{ classical field} \rightarrow (J) = \begin{array}{c} \text{---} \times \\ \text{---} \times \\ \text{---} \times \end{array} + \begin{array}{c} \text{---} \times \\ \text{---} \times \\ \text{---} \times \end{array} + \begin{array}{c} \text{---} \times \\ \text{---} \times \\ \text{---} \times \end{array} + \dots$$

which is just

$$[J] = Z[S] = \sum_{\{J\}} e^{iJ\phi}$$

$$\text{---} [J] = \text{---} (J) [J]$$

Define 1PI by a shaded blob

/// $\equiv$ 1PI

(i) line from i to a cross

(ii) Propagator from i to 1PI blob w one external pt.

(iii) prop. from i to 1PI w two external pts

$$(///) = \begin{array}{c} i \text{---} \times \\ \text{---} (///) \\ \text{---} (///) \text{---} (J) \end{array} + \frac{1}{2} \begin{array}{c} \text{---} (///) \text{---} (J) \\ \text{---} (///) \text{---} (J) \end{array} + \frac{1}{3!} \begin{array}{c} \text{---} (///) \text{---} (J) \\ \text{---} (///) \text{---} (J) \\ \text{---} (///) \text{---} (J) \end{array}$$

which is:

$$\phi_i^{cl} = \Delta_{ij} \left[1 - \left(\sum_j \Gamma_j^i + \prod_{jk} \phi_{jk}^{cl} + \frac{1}{2} \Gamma_{jke} \phi_{jk}^{cl} \phi_{ke}^{cl} + \dots + \frac{1}{(n-1)!} \Gamma_{j_1 \dots j_n} \phi_{j_1}^{cl} \dots \phi_{j_n}^{cl} + \dots \right) \right]$$

As such we need to define the generator of of 1PI Greens fns:

Generating functional of 1PI Green's fns:

$$\hat{\Gamma}(\phi^{cl}) = \int_i \phi_i^{cl} + \frac{1}{2} \Pi_{ij} \phi_i^{cl} \phi_j^{cl} + \dots$$

s.t.

$$\Gamma_{i_1 \dots i_n} = \frac{\delta^n}{\delta \phi_{i_1} \dots \delta \phi_{i_n}} \left(\hat{\Gamma}(\phi^{cl}) \right) \Big|_{\phi^{cl}=0}$$

Thus, $\phi_{i(n)}^{cl} = \Delta_{ij} J_j = - \frac{\delta W}{\delta J_i}$

Note that we must have different terms added for interactions

$$\phi_i^{cl} = \phi_0^{cl} + \phi_{int.}^{cl}$$

$$\begin{aligned} Z &= e^{-\int d^4x V(\frac{\delta}{\delta J})} e^{\frac{i}{2} \int d^4x \Delta^{-1} \phi^2} \\ &= e^{\frac{i}{2} \int d^4x \Delta^{-1} \phi^2} \left[e^{-\int d^4x V + J\phi} \right]_{\phi=0} \end{aligned}$$

$$S = S_0 + S_I[\phi] = J\phi$$

$$\Delta^{-1}_{ij} \phi_j + \delta_i S_I[\phi] = J_i = 0$$

$$\boxed{\phi_i^{cl} = \Delta_{ij} J_j - \Delta_{ij} (\delta_i S_I[\phi])}$$

Expand S_I around ϕ_0^{cl} :

$$S_I[\phi] = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{\delta^n S_I}{\delta \phi^n} \right) \phi_0^{cl}$$

$$= 1 + \frac{\delta S_I}{\delta \phi_i} \phi_0^{cl i} + \frac{1}{2} \frac{\delta^2 S_I}{\delta \phi_i \delta \phi_j} (\phi_0^{cl i} \phi_0^{cl j}) + \dots$$

$$\phi_i^{cl} = \Delta_{ij} \left(J_j - \frac{\delta S_D}{\delta \phi_j} \right)$$

$$[I[\phi] \equiv W[J_\phi] - \int d^d x \phi(x) J_\phi(x)]$$

$$\phi_i^{cl} = \Delta_{ij} \left(J_j - \frac{\delta \tilde{I}}{\delta \phi_j^{cl}} \right)$$

$$\text{or, } -\frac{\delta W}{\delta J_i} = \phi_i^{cl} = \Delta_{ij} \left(J_j - \frac{\delta \tilde{I}}{\delta \phi_j^{cl}} \right)$$

$$\Delta^{-1}_{ij} \phi_j^{cl} + \frac{\delta \tilde{I}}{\delta \phi_i^{cl}} = J_i$$

$$\frac{\delta}{\delta \phi_i^{cl}} \left[\tilde{I} + \frac{1}{2} \phi_i^{cl} \Delta^{-1}_{ij} \phi_j^{cl} \right] = J_i$$

We define the effective Action as

$$\Gamma(\phi^{cl}) \equiv \tilde{I}(\phi^{cl}) + \frac{1}{2} \phi_i^{cl} \Delta^{-1}_{ij} \phi_j^{cl}$$

$$\text{Such that } \frac{\delta \Gamma}{\delta \phi_i^{cl}} = J_i$$

which minimizes the Classical E.O.M :

$$\delta S / \delta \phi_i = J_i$$

* ϕ^{cl} plays the role of the field ϕ in the classical field theory

$$\phi \rightarrow \phi^{cl}$$

$$S \rightarrow \Gamma$$

$$\Gamma(\phi^{cl}) = \sum_{n=1}^{\infty} \frac{1}{n!} \Gamma_{i_1 \dots i_n} \phi_{i_1}^{cl} \dots \phi_{i_n}^{cl}$$

$$\Gamma_{ij} = \Pi_{ij} + \Delta_{ij}^{-1} \quad \Rightarrow \quad \textcircled{2/11} + (\text{---})^{-1}$$

Aside on what's going on

$$\langle S_i(\tilde{\phi}) \rangle_{\tilde{J}} = \tilde{J}_i$$

$$Z[\tilde{J}] = \int D\phi \, e^{-S_0 + \tilde{J}\phi}$$

critical pt of $\frac{\delta S}{\delta \phi_i} = \tilde{J}_i$
instead look at average of this eq:

$$\langle \delta_i S \rangle = \tilde{J}_i$$

Note that

$$A(\phi) = \sum_{n=0}^{\infty} \frac{1}{n!} A_{i_1 \dots i_n} \phi^{i_1} \dots \phi^{i_n}$$

$$\langle A(\phi) \rangle = \sum_{n=0}^{\infty} \frac{1}{n!} A_{i_1 \dots i_n} \langle \phi^{i_1} \dots \phi^{i_n} \rangle$$

$$G_{i_1 \dots i_n} = \langle 0 | \phi_{i_1} \dots \phi_{i_n} | 0 \rangle = \left(\frac{\delta}{\delta J_{i_1}} \dots \frac{\delta}{\delta J_{i_n}} \right) \frac{Z|_{J=0}}{Z}$$

$$\langle A(\phi) \rangle = \left(\sum_{n=0}^{\infty} \frac{1}{n!} A_{i_1 \dots i_n} \frac{\delta}{\delta J_{i_1}} \dots \frac{\delta}{\delta J_{i_n}} \right) \frac{Z[J]}{Z[J]} \Big|_{J=0}$$

$$= \frac{\hat{A}(\frac{\delta}{\delta J}) Z[J]}{Z[J]} \Big|_{J=0}$$

Proof:

Thus,

$$Z[J]$$

$$\langle 4 \rangle = \frac{\delta \langle S \rangle}{\delta J} = \frac{\delta}{\delta J} \langle S + J \rangle$$

$$\left\{ \begin{array}{l} \frac{\delta \langle S \rangle}{\delta J} \langle 4 \rangle = \langle S + J \rangle = 0 \Rightarrow \langle S \rangle = -J \end{array} \right.$$

$$\langle \delta_i S \rangle = -J_i$$

$$\left\{ \delta_i S \left(\frac{\delta}{\delta J} \right) \right\} (Z[J]) \Big|_{J=0} = Z[J] J_i$$

which is a differential eq. for $Z[J]$:

$$\tilde{D}_i \equiv \delta_i S \left(\frac{\delta}{\delta J} \right)$$

$$(\tilde{D}_i + J_i) Z[J] = 0$$

We solve via F.T.:

Conjugate variable to J is ϕ :

$$Z[J] = \int D\phi f(\phi) e^{-J\phi}$$

$$\left(\delta_i S \left(\frac{\delta}{\delta J} \right) + J_i \right) \int D\phi f(\phi) e^{-J\phi} = 0$$

$$= \int D\phi f(\phi) (\delta_i S(\phi) + J_i) e^{-J\phi} = 0$$

$$= \int D\phi f(\phi) \left(\frac{\delta S}{\delta \phi_i}(\phi) - \frac{\delta}{\delta \phi_i} \right) e^{-J\phi} = 0$$

$$= \int D\phi \left(\frac{\delta S}{\delta \phi_i} + \frac{\delta}{\delta \phi_i} \right) f(\phi) e^{-J\phi} \quad \text{int. by parts}$$

$\therefore$ We obtain the following Expression

$$\frac{\delta f}{\delta \phi_i} = -f(\phi) \frac{\delta S}{\delta \phi_i}$$

or (formally)

$$\delta f = -f(\phi) \delta S \Rightarrow \ln(f(\phi)) = -S[\phi]$$

$$f(\phi) = e^{-S[\phi]}$$

...

This was useless... lol.

Thm: The effective Action is the Legendre transform of the free energy:

$$e^{-\Gamma(\phi^{cl}) + \int J_i \phi_i^{cl}} = \int D\phi e^{-S[\phi] + \int J_i \phi_i} = e^{-W[J]}$$

$$\Gamma[\phi^{cl}] = W[J] + \int J_i \phi_i^{cl}$$

Proof: $\frac{\delta \Gamma[\phi^{cl}]}{\delta \phi_i^{cl}} = J_i$

but

$$\frac{\delta W}{\delta \phi} = \frac{\delta J}{\delta \phi} \frac{\delta W}{\delta J} = \frac{\delta J}{\delta \phi} (1 - \phi^{cl})$$

$$+ \frac{\delta J}{\delta \phi} \phi_{cl} \leftarrow \text{terms cancel.} \quad \square$$

$$\langle \phi \rangle = \frac{\partial}{\partial J} \frac{Z}{Z} = - \frac{\partial W}{\partial J} \frac{Z}{Z}$$

$$\langle \phi \rangle = - \partial W / \partial J = \phi^{cl}$$

The Connected two pt fn:

$$G_c^i[J] \equiv - \frac{\delta W[J]}{\delta J_i} = \phi_i^{cl}[J]$$

We take the derivative of this expression

$$G_{ij}^c[J] = - \frac{\delta W[J]}{\delta J_i \delta J_j} = \frac{\delta \phi_i^{cl}}{\delta J_j}$$

But recall $\phi_i^{cl} = \Delta_{ij} (J_j - (I_j + \dots))$

$$\frac{\delta \phi_i^{cl}}{\delta J_k} = \Delta_{ik} - \Delta_{ik} \Pi_{ne} \frac{\delta \phi_e^{cl}}{\delta J_k} + \dots$$

$$\therefore G_{ij}^C[J] = \Delta_{ij} - \Delta_{ik} \Pi_{kl} \frac{\delta \phi_l^C}{\delta J_i} + \dots$$

Assume: $\phi_i^C[J=0] = 0$

↳ Since we are left with $\Delta_{ij}^{-1} \phi = 0$

At $J=0$:

$$G_{ij}^C[0] = \Delta_{ij} - \Delta_{ik} \Pi_{kl} \frac{\delta \phi_l^C}{\delta J_i}$$

$\leftarrow G_{lj}^C$

$$(\Delta_{ik} \Pi_{kl} + \delta_{li}) G_{lj} = \Delta_{ij}$$

$$(\Delta_{im}^{-1} \Delta_{ik} \Pi_{kl} + \Delta_{lm}^{-1}) G_{lj} = \delta_{mj}$$

$$(\Pi_{ml} + \Delta_{ml}^{-1}) G_{lj} = \delta_{mj}$$

$$(\Pi_{ml} + \Delta_{ml}^{-1}) G_{lj} = \delta_{mj}$$

* The Connected 2pt Γ^C is the inverse of the 2pt Γ

$$\Gamma_{ik} G_{kj}^C = \delta_{ij}$$

Formally, $(\Delta \Pi + 1) G^C = 1$

$$\therefore G^C = (1 + \Delta \Pi)^{-1} \Delta$$

Expanding this expression we find:

$$(1 + \Delta \Pi)^{-1} = (1 - \Delta \Pi + (\Delta \Pi)^2 - \dots)$$

$$G^C = (1 - \Delta \Pi + \Delta \Pi \Delta \Pi + \dots) \Delta$$

$$= (\Delta - \Delta \Pi \Delta - \Delta \Pi \Delta \Pi \Delta + \dots)$$

$$G_{ij}^C = (\Delta_{ij} - \Delta_{ik} \Pi_{kl} \Delta_{lj} - \Delta \dots)$$

$$G^C = \dots$$

$$a \rightarrow (c) \rightarrow b \Rightarrow a \rightarrow b - a \rightarrow (c) \rightarrow b + (c) \rightarrow (c) \rightarrow b$$

$$\star \delta \left[\int d^4x \left(-\partial_\mu \phi^2 - \frac{1}{2}(m^2 - i\varepsilon) \phi^2 + \mathcal{L} \phi \right) \right] \\ = \int d^4x \left(\square \phi - \tilde{m}^2 \phi + \mathcal{L} \right) \delta \phi$$
$$(-\square + \tilde{m}^2) \phi_J = J$$

$$(-\Box + m^2) \equiv \Delta$$

$$\leadsto \phi_J = \hat{\Omega}^{-1} J$$

$$\begin{aligned} (p^2 + m^2) \phi_J &= \int d^4x' J(x') e^{-ipx'} \\ \phi_J(p) &= \left(\frac{1}{p^2 + m^2} \right) \int d^4x' J(x') e^{-ipx'} \end{aligned}$$

$$\phi_J(p) = \left(\frac{1}{p^2 + \vec{u}^2} \right) \int d^4x \, \delta(x) e^{-i p x}$$

$$\phi_J(x) = \int \frac{d^d p}{(2\pi)^d} \int d^d x' \frac{J(x')}{(p^2 + m^2)} e^{-i p x'} e^{i p x}$$

$$\phi_J(k) = \int \frac{d^d p \, d^d x'}{(2\pi)^d} \left(\frac{J(x')}{p^2 + k^2} \right) e^{i p(x - x')}$$

$$\bullet Z[J] = \int D\phi e^{-S_0 + iJ\phi}$$

We can make a change of variables

$$i_5 = 0$$

$$\Phi = \Phi_{\text{constant}} + \chi(x)$$

$$\int \mathcal{D}\phi = \int \mathcal{D}\chi$$

$$: (10x)^2 - 1(7d)^2 = 71^2 - 1(7^2)(14 + 14)$$

$$\therefore Z[J] = \int \mathcal{D}\chi \, e^{i \int d^4x \left[\frac{1}{2} (\partial_\mu \chi + \partial_\mu \phi) - \frac{1}{2} m^2 (\phi + \chi)^2 + J(\phi + \chi) \right]}$$

Action:

$$\begin{aligned} & -\frac{1}{2} \partial_\mu (\phi + \chi) \partial^\mu (\phi + \chi) - \frac{1}{2} m^2 (\phi + \chi)^2 + J(\phi + \chi) \\ &= \frac{1}{2} (\phi + \chi) \partial^2 (\phi + \chi) - \frac{1}{2} m^2 (\phi + \chi)^2 + J(\phi + \chi) \\ &= \left(\frac{1}{2} (\phi + \chi) \right) \left[\partial^2 - m^2 \right] (\phi + \chi) + J(\phi + \chi) \\ &= -\frac{1}{2} (\phi + \chi) \hat{\square} (\phi + \chi) + J(\phi + \chi) \\ &= -\frac{1}{2} \phi \hat{\square} \phi - \frac{1}{2} (\underbrace{\chi \hat{\square} \phi + \phi \hat{\square} \chi}_{\text{cross terms}}) - \frac{1}{2} \chi \hat{\square} \chi \\ &= \underbrace{+ J\phi}_{\text{cross terms}} + J\chi \\ &= -\frac{1}{2} \phi \hat{\square} \phi - \frac{1}{2} \chi \hat{\square} \chi + J\chi \end{aligned}$$

(c) Cross terms:

$$\begin{aligned} & -\frac{1}{2} \phi \hat{\square} \chi - \frac{1}{2} \partial_\mu \chi \partial^\mu \phi - m^2 \phi \chi + J\chi \\ &= \chi \partial^2 \phi - m^2 \phi \chi + J\chi \\ &= \chi (\underbrace{\square - m^2 + J}_{=0}) \phi \end{aligned}$$

Thus

$$Z[J] = e^{-\frac{1}{2} \phi \hat{\square} \phi} \int \mathcal{D}\chi \, e^{-\frac{1}{2} \chi \hat{\square} \chi}$$

$$\therefore \boxed{Z[J]/Z[0] = e^{-\frac{1}{2} J \tilde{\square}^{-1} J}}$$

Define: $G_F(x, y) = -i \tilde{\square}^{-1} \delta^4(x - y)$

Thus

$$= \frac{1}{2} \int d^4x \int d^4y J_x G_{xy} J_y$$

$$Z[\delta J] / Z[0] = e$$

Since $\int d^d x J(x) \left(\frac{-i}{\Box_x + m^2} \right) J(x)$
 $J(x) = \int d^d y \delta^d(x-y) J(y)$
 and so

$$\int d^d x \int d^d y J(x) G(x,y) J(y)$$

$$Z_0[J] \equiv \int D\phi e^{iS_{\text{free}} + J\phi}$$

$$Z_0[0] = \int D\phi e^{iS_{\text{free}}}$$

$$\langle \Omega | e^{i \int d^d x J\phi} | \Omega \rangle = \frac{Z_0[J]}{Z_0[0]} = e^{\frac{1}{2} J \Delta J}$$

* $S = \int d^d x \left(-\frac{1}{2} (\partial\phi)^2 - \frac{1}{2} \tilde{m}^2 \phi^2 + \mathcal{L}_{\text{int}}(\phi) \right)$

The new partition fn

$$Z[J] = \langle \Omega | e^{\int d^d x \mathcal{L}_{\text{int}}(\phi)} e^{iJ\phi} | \Omega \rangle$$

which we related to $Z_0[J]$ earlier.

$$Z[J] = \int D\phi e^{i \int d^d x \left(-\frac{1}{2} \partial\phi^2 - \frac{1}{2} \tilde{m}^2 \phi^2 + J\phi + \mathcal{L}_{\text{int}} \right)}$$

$$Z[J] = \int D\phi \left(\sum_{n=0}^{\infty} \frac{i^n}{n!} \left(\int d^d x \mathcal{L}_{\text{int}} \right)^n e^{iS_0[\phi, J]} \right)$$

$$= \sum_{n=0}^{\infty} \frac{i^n}{n!} \int D\phi \left(\mathcal{L}_{\text{int}}^n \right) e^{iS_0[\phi, J]}$$

$$= \sum_{n=0}^{\infty} \frac{i^n}{n!} Z_n[J]$$

where we have defined

$$Z_n[J] \equiv \prod_{p=1}^n \left(\int d^d x_p \right) \int D\phi \left(\prod_{p=1}^n \mathcal{L}(\phi_p) \right) e^{iS_0[\phi, J]}$$

$$= \frac{1}{n!} \left(\int d^d x_p \right) \int D\phi \frac{1}{n!} \mathcal{L} \left(\frac{1}{n} \frac{\delta}{\delta J_p} \right) e^{iS_0[\phi, J]}$$

$$= \left(\frac{1}{P-1} \int d^d x_p \mathcal{L} \left(\frac{1}{i} \frac{\delta}{\delta \phi_p} \right) \right)^P \int D\phi e^{iS[\phi-J]}$$

$$\therefore Z[J] = e^{i \int d^d x \mathcal{L} \left(\frac{1}{i} \frac{\delta}{\delta \phi} \right)} Z_0[J]$$

$$Z[J] = Z_0[0] e^{i \int d^d x \mathcal{L} \left(\frac{1}{i} \frac{\delta}{\delta \phi} \right)} e^{\frac{1}{2} J \Delta J}$$

$\lambda \phi^4$ Example: $-\frac{1}{4} \lambda \phi^4$

$$Z[J] = e^{i \int d^d x \lambda \frac{\delta^4}{4!}} e^{-\frac{1}{2} J \Delta J}$$

$$\left(1 + i\lambda + i^2 \lambda^2 / 2! + \dots \right) e^{-\frac{1}{2} J \Delta J}$$

$$\underline{\lambda} \therefore i\lambda \left(\frac{\delta^4}{\delta J^4} e^{-\frac{1}{2} J \Delta J} \right)$$

$$= i\lambda \delta_x^3 (-\Delta_{x\omega} J^\omega) e$$

$$= i\lambda \delta_x^2 (\Delta_{xx} - (\Delta_{x\omega} J^\omega)^2) e$$

$$= (-i\lambda) \delta_x \left[(-2(\Delta_{xx}) \Delta_{x\omega} J^\omega) e - \Delta_{xx} \Delta_{x\omega} J^\omega e + (\Delta_{x\omega} J^\omega)^3 e \right]$$

$$= -i\lambda \delta_x [+3 \Delta_{xx} \Delta_{x\omega} J^\omega - (\Delta_{x\omega} J^\omega)^3] e$$

$$= \frac{-i\lambda}{4!} [3\Delta_{xx}^2 - 3\Delta_{xx} (\Delta_{x\omega} J^\omega)^2 - 3\Delta_{xx} (\Delta_{x\omega} J^\omega)^2 + (\Delta_{x\omega} J^\omega)^4] e$$

$$= \frac{-i\lambda}{4!} [3\Delta_{xx}^2 - 6 \Delta_{xx} (\Delta_{x\omega} J^\omega)^2 + (\Delta_{x\omega} J^\omega)^4] e$$

$$= -\frac{i\lambda}{8} \Delta_{xx}^2 + \frac{i\lambda}{4} \Delta_{xx} (\Delta_{x\omega} J^\omega)^2 - \frac{i\lambda}{4!} (\Delta_{x\omega} J^\omega)^4$$

$$\frac{-i\lambda}{8} \text{ (loop diagram)} + \frac{i\lambda}{4} \text{ (tadpole diagram)} - \frac{i\lambda}{4!} \text{ (4-point vertex diagram)}$$

$$\begin{aligned}
Z[0] &= \int D\phi e^{iS_0 + iS_{int}} \\
&= \int D\phi e^{iS_0} (1 + i\lambda/\phi! \phi^4 + \dots) \\
&= 1 - \frac{i\lambda}{4!} \langle \phi_x \phi_x \phi_x \phi_x \rangle + \dots \\
&= 1 - \frac{i\lambda^3}{4!} \Delta_{xx} \Delta_{xx} + \mathcal{O}(\lambda^2) \\
&= 1 - \frac{i\lambda}{8} \bigcirc
\end{aligned}$$

We now divide by this factor:

$$\begin{aligned}
Z[J]/Z[0] &= (1 - \alpha)/(1 - \beta \cdot) e^{-\frac{1}{2}J\Delta J} \\
(1 - \beta)^{-1} &\approx 1 + \beta
\end{aligned}$$

and so to the same order in λ

$$\begin{aligned}
Z[J]/Z[0] &= e^{-\frac{1}{2}J\Delta J} \left(1 - \frac{i\lambda}{8} \bigcirc + \frac{i\lambda}{4} \times \Delta \times - \frac{i\lambda}{4!} \times \times \times \right. \\
&\quad \left. + \frac{i\lambda}{8} \bigcirc \right)
\end{aligned}$$

$$\frac{Z[J]}{Z[0]} = Z_0[J] (\text{Non-vacuum bubbles})$$